



* JURUS *

CRASH COURSE



Memperkasa Kesiapsiagaan Akar Umbi untuk Ketahanan Siber Negara melalui Kompetensi Praktikal Berdasarkan Situasi Sebenar dalam Industri



SILIBUS JURUS

“Framework” Kompetensi & Standard Penilaian

FOKUS UTAMA



ANALYST
Foundational Operator

Pembinaan kerangka asas, konfigurasi yang selamat dan kesiap-sediaan membaik pulih system dalam satu hos

PENILAIAN MODUL

- Kejuruteraan Sistem
- Rangkaian dan Keselamatan
- Keselamatan Pangkalan Data
- Keselamatan Aplikasi
- Pemantauan Keselamatan
- Kesenambungan Perniagaan / Kelangsungan Operasi Perniagaan

KAEDAH PENILAIAN

- “LIVE-DEMO” dengan sesi walkthrough
- Senarai semak berasaskan bukti (Screenshot/Log)
- Audit hardening
- Benchmark Pembentangan lisan langkah penyelesaian masalah
- Pengukuran MTD/MTO dalam kesinambungan perniagaan

HASIL AKHIR

Mampu untuk membina satu platform yang **“SAFE & SECURE”** secara individu



PROFESSIONAL
Integration Engineer

Integrasi sistem selamat, segmentasi rangkaian dan keupayaan pemulihan merentas persekitaran multi-tier.

- Deployment Seni Bina Multi-Tier
- Segmentasi Rangkaian & Kawalan Akses
- Seni Bina VPC Cloud
- Integrasi Aplikasi-Pangkalan Data
- Skrip Automasi Asas

- Pembinaan senario + change ticket
- Pemulihan daripada dapatan audit simulasi
- Simulasi pemulihan Blue Team
- Bukti prinsip Least Privilege & Logging

Keupayaan mengintegrasikan sistem secara **selamat** dan memulihkan isu.



SPECIALIST
Resilience Architect

Reka bentuk ketersediaan tinggi, automasi lanjutan berskala besar, seni bina secure-by-default dan pembinaan semula berasaskan insiden.

- Corak HA/DR
- Seni Bina Awan Lanjutan (Multi-Region)
- Automasi Hardening (Ansible/Terraform)
- DevSecOps & Security Gate CI/CD
- Pembinaan Semula Infrastruktur Pasca Pencerobohan
- Penyelesaian Masalah Insiden Kompleks

- Simulasi Red-Blue Team (Live Fire)
- Pengukuran Service Level Indicator(SLI)/Service Level Objective(SLO) seperti dalam Site Reliability Engineering (SRE) Google
- Architecture Review Board Defence
- Analisis Trade-off Kos-Risiko

Keupayaan mereka bentuk dan **mempertahankan seni bina** yang tahan lasak.



SECURITY FRAMEWORK SUMMARY – 6 KEY DOMAINS OVERVIEW

1

SYSTEM ENGINEERING



2

NETWORK SECURITY



3

DATABASE & DATA SECURITY



4

APPLICATION SECURITY



5

SECURITY MANAGEMENT



6

BUSINESS RESILIENCY



6
DOMAINS



30+
KEY AREAS



100+
CONSIDERATIONS



RESILIENT
BY DESIGN



RISK-AWARE
& PROACTIVE



SECURE TODAY
RESILIENT TOMORROW

1. SYSTEM ENGINEERING – KEY AREAS OVERVIEW

KEY AREA	EXAMPLES / CONSIDERATIONS			
 1. HARDWARE & SOFTWARE (PHYSICAL, ON PREM, VIRTUAL, CLOUD)	 Physical	 On Prem	 Virtual	 Cloud
 2. ENVIRONMENTAL & PHYSICAL SECURITY (PHYSICAL, ENERGY...)	 Physical Security	 Power & Energy	 Environment (Temp, Humidity)	 Fire Protection
 3. OPERATING SYSTEMS (WINDOWS, LINUX, KUBERNETES, SERVERLESS...)	 Windows	 Linux	 Kubernetes	 Serverless
 4. IDENTITY & ACCESS MANAGEMENT (LOCAL/LDAP/AD)	 Local	 LDAP	 Active Directory (AD)	 RBAC / Least Privilege
 5. CHANGE, CONFIGURATION, RELEASE & PATCH MANAGEMENT	 Change Management	→  Configuration Management	→  Release Management	→  Patch Management
 6. HIGH LEVEL DESIGN	 Requirements	→  Architecture	→  Components	→  Design Principles
 7. NETWORK ARCHITECTURE DESIGN (RESILIENCY)	 Multi-Region	 Load Balancing	 Redundancy / Failover	 High Availability

2. NETWORK SECURITY – KEY AREAS OVERVIEW

KEY AREA	EXAMPLES / CONSIDERATIONS
 1. NETWORK SEGMENTATION (EXTERNAL, DMZ, INTERNAL)	 External (Internet)  DMZ (Public Services)  Internal (Private Network)
 2. NETWORK PERIMETER SECURITY (DEFENSES IN DEPTH)	 Firewall  IDS/IPS  Web Filtering  Email Security  DDoS Protection
 3. REMOTE ACCESS SECURITY	 VPN  MFA  Device Posture  Secure Gateway
 4. NETWORK ACCESS CONTROL	 Identity Verification  Device Compliance  Role-Based Access  Least Privilege
 5. NETWORK MONITORING	 Traffic Visibility  Anomaly Detection  Alerting  Reporting & Forensics

3. DATABASE & DATA SECURITY – KEY AREAS OVERVIEW

KEY AREA	EXAMPLES / CONSIDERATIONS			
 1. DATABASE DESIGN	 Data Modeling	 Keys & Relationships	 Normalization	 Performance Optimization
 2. DATABASE SECURITY	 Access Control	 Authentication & Authorization	 Auditing & Compliance	 Encryption (in Transit)
 3. DATA PROTECTION	 Data Encryption	 Backup & Recovery	 Data Masking	 Data Retention
 4. TRANSACTION LOG SHIPPING	 Transaction Logs	 Log Shipping	 Standby Database	 Disaster Recovery
 5. DATABASE MONITORING	 Performance Monitoring	 Anomaly Detection	 Alerts & Notifications	 Health Checks

4. APPLICATION SECURITY – KEY AREAS OVERVIEW

KEY AREA

EXAMPLES / CONSIDERATIONS



1. WEB / MOBILE APPLICATION DESIGN



Secure Architecture



Security by Design



Secure Mobile



Usability & Privacy



2. API DESIGN



REST / GraphQL



Authentication & Authorization



Input Validation



Rate Limiting & Throttling



3. IAAA (IDENTITY, AUTHENTICATION, AUTHORIZATION & ACCOUNTING)



Identity Management



Strong Authentication



Authorization (RBAC/ABAC)



Audit & Accounting



4. SDLC PROCESS



Requirements & Analysis



Secure Development



Security Testing



Deployment



Maintenance



5. APPLICATION VULNERABILITY MANAGEMENT



Vulnerability Scanning



Risk Assessment



Remediation & Patch



Tracking & Reporting

5. SECURITY MANAGEMENT – KEY AREAS OVERVIEW

KEY AREA	EXAMPLES / CONSIDERATIONS			
 1. LOG MANAGEMENT DESIGN	 Log Sources	 Collection & Ingestion	 Storage & Retention	 Integrity & Protection
 2. LOG ANALYSIS AND CORRELATION	 Search & Investigation	 Correlation Rules	 Anomaly Detection	 Alert Generation
 3. SECURITY OPERATION CENTER	 24/7 Monitoring	 Triage & Escalation	 Dashboards & Visibility	 Threat Intelligence
 4. INCIDENT MANAGEMENT	 Detection & Validation	 Classification & Prioritization	 Assignment & Coordination	 Resolution & Closure
 5. INCIDENT REPORT	 Executive Summary	 Timeline & Impact	 Root Cause Analysis	 Lessons Learned
 6. ADVANCE INCIDENT RESPONSE (THREAT HUNTING & COMPROMISE ASSESSMENT)	 Threat Hunting	 Compromise Assessment	 Containment & Eradication	 Continuous Improvement

6. BUSINESS RESILIENCY – KEY AREAS OVERVIEW

KEY AREA	EXAMPLES / CONSIDERATIONS			
 1. BUSINESS CONTINUITY MANAGEMENT (BCM)	 Business Impact Analysis	 Continuity Strategy	 Plan, Test & Maintain	 Roles & Communication
 2. DISASTER RECOVERY PLAN (DRP)	 Recovery Strategy	 Recovery Site	 RTO / RPO Targets	 DR Testing & Validation
 3. HIGH AVAILABILITY DESIGN (HA)	 Redundancy & Failover	 Load Balancing	 Clustering & Replication	 Health Monitoring
 4. BACKUP DESIGN	 Backup Strategy	 Backup Storage	 Backup Schedule	 Backup Testing & Validation
 5. INCIDENT MANAGEMENT (FOCUS HIGH IMPACT)	 High Impact Detection	 Rapid Response & Escalation	 Impact Mitigation	 Post-Incident Review



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